

SMART PLANT POT SYSTEM WITH SOIL MOISTURE MONITORING AND ALERT INTERFACE

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ABSTRACT

This paper presents the design and implementation of a smart plant pot system based on the ESP32 microcontroller, a 2.4-inch TFT display, and a soil moisture sensor. The system continuously monitors soil moisture, displays data through a graphical user interface (GUI) built with LVGL, and issues alerts when soil moisture exceeds safe thresholds (too dry or too wet). The solution targets home and office users, enabling more accurate plant care without agricultural expertise. Test results show that the system is stable, with an intuitive interface and timely alerts.

Keywords: Smart plant pot, ESP32, soil moisture, TFT display, LVGL, IoT, alert.

1 INTRODUCTION

1.1 Background and Motivation

Indoor and office gardening has become increasingly popular due to benefits in aesthetics, air quality, and health. However, under- or over-watering often leads to wilting or root rot. Many people lack the experience or time to monitor soil moisture regularly. Smart plant pot systems have emerged to automate monitoring and support irrigation decisions through sensors and user interfaces.

1.2 Objectives

- **Main objectives:** To build a smart plant pot system using ESP32, a 2.4-inch TFT display, and a soil moisture sensor to:
 1. Measure and display soil moisture in real time on a graphical interface.
 2. Alert when soil moisture is at a dangerous level (too low—need to water, or too high—risk of waterlogging).

3. Provide a user-friendly, readable interface (numeric value, % unit, and optionally time, temperature, and air humidity if extended).
- **Secondary objectives:** To serve as a platform for extending features (automatic watering, WiFi/mobile app connectivity, data history).

1.3 Scope and Target Users

- **Scope:** One plant pot, one soil moisture sensor, one directly attached TFT display (standalone); internet connection is optional in the basic version.
- **Target users:** Home and office users, beginners in plant care.

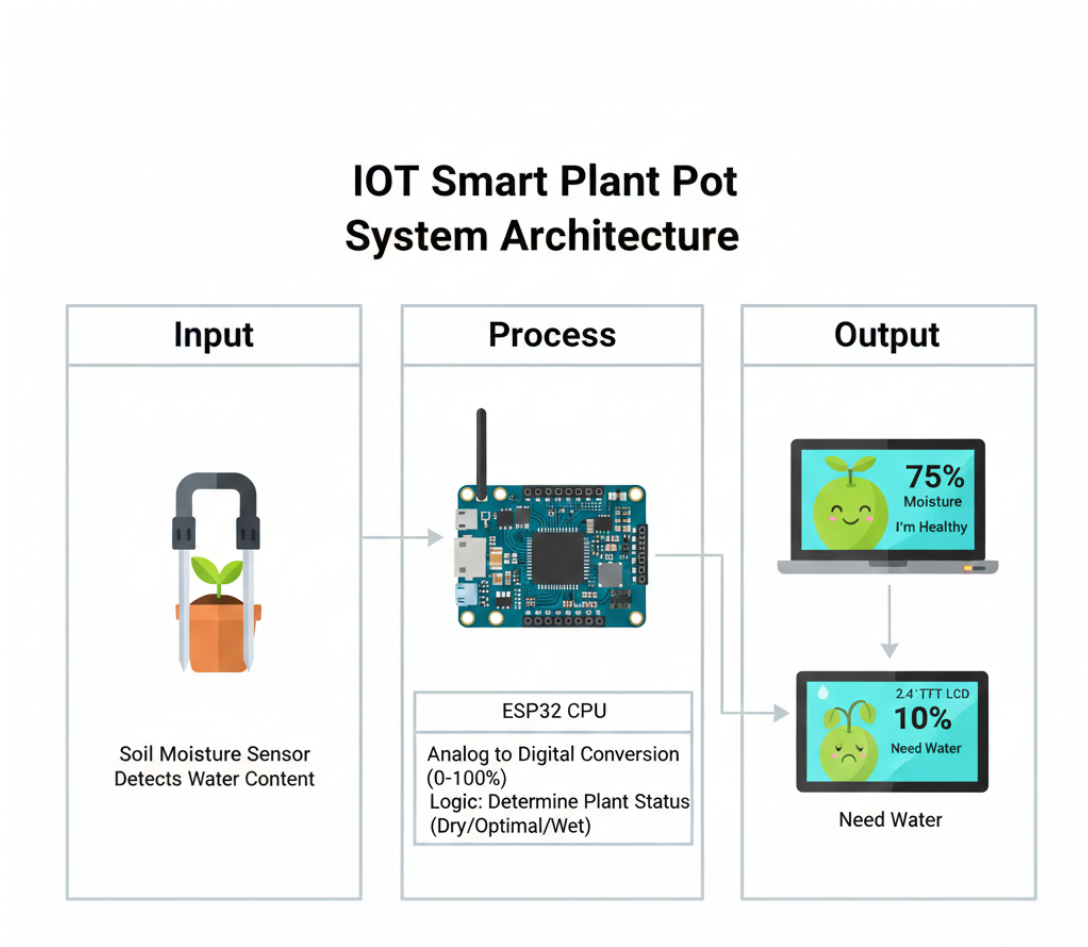


Figure 1: System architecture: sensor → ADC → ESP32 → display and alerts.

2 THEORETICAL BACKGROUND AND RELATED TECHNOLOGY

2.1 Soil Moisture and Its Effect on Plants

Soil moisture is commonly expressed as a percentage (%) of volumetric or gravimetric moisture content. For potted plants, volumetric percentage is intuitive and compatible with many commercial sensors.

Safe thresholds (depending on plant type, typically):

- **Too dry (danger):** moisture $< 20\text{--}30\%$ $\rightarrow$ need to water.
- **Too wet (danger):** moisture $> 80\text{--}90\%$ $\rightarrow$ risk of waterlogging and root rot.
- **Stable:** approximately $40\text{--}70\%$ for many indoor plants.

Specific thresholds can be configured in software for each plant type.

2.2 Soil Moisture Sensor

Soil moisture sensors typically use **resistive** or **capacitive** principles. Capacitive sensors are less prone to corrosion and are suitable for long-term use. Output is usually **analog (voltage)** or **digital (HIGH/LOW)**. The ESP32 has an ADC, making it easy to read analog values and convert them to moisture percentage (after calibration).

2.3 ESP32 Microcontroller

The **ESP32** (Espressif) is a 32-bit microcontroller with WiFi/Bluetooth, multiple GPIO pins, ADC, and DAC, suitable for IoT and local display applications. The **Arduino framework** on ESP32 enables rapid development with rich libraries (TFT, LVGL, ADC, etc.).

2.4 TFT Display and LVGL

The **2.4-inch TFT** (240×320 pixels) is an SPI color display that saves pin count and is suitable for embedded products. **TFT_eSPI** is a popular TFT driver library for ESP32, optimized for drawing speed. **LVGL (Light and Versatile Graphics Library)** is an open-source GUI library supporting widgets (label, panel, bar, image, etc.), suitable for building professional GUIs on embedded displays. Version 8.3 is used in this project.

3 SYSTEM DESIGN

3.1 Overall Architecture

The system consists of three main layers:

1. **Sensor layer:** Soil moisture sensor placed in the soil; signal fed to the ESP32 ADC pin.

2. **Processing layer:** ESP32 reads the ADC, computes moisture (%), compares with alert thresholds, and updates the interface data.
3. **Display and alert layer:** The 2.4-inch TFT displays the GUI (LVGL); alerts can be shown by color (red/yellow/green), text (“Need water” / “OK” / “Too wet”), and optionally LED or buzzer.



3.2 Hardware

Table 1: Hardware components

Component	Description / Specification
Microcontroller	ESP32 (DOIT DevKit v1)
Display	TFT 2.4 inch, 240×320, SPI interface
Soil moisture sensor	Resistive/capacitive, analog output (0–3.3 V)
Power supply	USB 5 V or 5 V adapter
Display backlight	PWM control (GPIO 25)

Connections: Sensor to ADC pin (e.g., GPIO 34); display wired per TFT_eSPI configuration (SPI: CLK, MOSI, MISO, CS, DC, RST, backlight).

3.3 Software

- **Environment:** PlatformIO, Arduino framework, ESP32 DOIT DevKit v1 board.
- **Main libraries:**
 - **lvgl** (8.3.11): GUI (main screen, moisture panel, value/unit labels, clock, date, etc.).
 - **TFT_eSPI** (2.5.43): display driver, provides flush callback for LVGL.
- **Interface:** Designed with SquareLine Studio; C code exported and integrated into the project.
- **Processing flow:**
 1. Initialization: TFT, LVGL, load UI (`ui_init`).
 2. Main loop: read ADC → convert to moisture % → update moisture label on LVGL → compare with thresholds → update alert state → call `lv_timer_handler()` and delay 10 ms.

3.4 Alert Thresholds and Display

- **Thresholds (configurable in code):**
 - Dry (need water): moisture $< 30\%$.
 - Normal: $30\% \leq \text{moisture} \leq 70\%$.
 - Too wet: moisture $> 70\%$.
- **GUI display:** Moisture panel border color: red (dry), green (normal), yellow/orange (too wet). Text: “Need water” / “OK” / “Too wet”. Optionally LED or buzzer when dry or too wet.

4 TECHNICAL IMPLEMENTATION

4.1 Hardware and Display Driver Configuration

The `config.h` file defines: display size 240×320 ; LVGL draw buffer (e.g., `screenWidth * 10` pixels); `my_disp_flush` calls `TFT_eSPI (setAddrWindow, pushColors)` to draw the region requested by LVGL, then calls `lv_disp_flush_ready`; TFT initialization (`init`, `setRotation`, `fillScreen`) and backlight PWM.

4.2 LVGL and SquareLine Integration

`lvgl_init()`: Initializes LVGL, draw buffer, and registers the display driver (`hor_res`, `ver_res`, `flush_cb`, `draw_buf`). `ui_init()`: Initializes theme, creates main screen (`Screen1`), loads SquareLine components (`BackGround`, `PanelSoilMoisture`, `MoistureValue1`, `MoistureUnit`, `clock`, `date`, etc.) and loads the screen onto the display.

4.3 Sensor Reading and Interface Update

Read ADC from the pin connected to the soil moisture sensor. Map ADC value (min–max) to 0–100% (calibration with actual soil is required). Use `lv_label_set_text_fmt` to update `ui_MoistureValue1` with the percentage. Compare with thresholds; change style of `ui_PanelSoilMoisture` (`border_color`, `border_width`) and update status label.

4.4 Alert on Dangerous Moisture Level

In the main loop: if $< 30\%$: set “dry” alert style (red), optionally turn on LED/buzzer; if $30\text{--}70\%$: set “normal” style (green); if $> 70\%$: set “too wet” style (yellow/orange). To avoid flicker when the value oscillates near the threshold, add hysteresis (e.g., switch to “dry” only when $< 25\%$, back to “normal” when $> 35\%$).

5 RESULTS AND DISCUSSION

5.1 Display Interface

The main interface includes: graphical background, **soil moisture panel** with numeric value and % unit, clock (hour : minute : second), date and weekday, and additional

info (temperature, air humidity, PM2.5, wind speed, rain, etc.) as designed in SquareLine Studio. Soil moisture is updated in real time, is easy to read, and is suitable for placement next to the plant pot.

5.2 Alerts

Changing the panel border color and status text helps users quickly recognize when the soil is too dry or too wet. The system detects and switches alert state within a few seconds after moisture crosses the threshold.

5.3 Limitations and Future Work

- **Calibration:** The ADC–moisture % relationship depends on soil type and sensor; calibration is needed for each environment.
- **ADC noise:** Noise can be reduced by averaging multiple readings or a low-pass filter.
- **Extensions:** Add automatic watering module (relay + pump), log moisture history, WiFi connectivity for alerts to phone or web dashboard.



Figure 2: Smart plant pot system in use: display and sensor beside the plant.

6 CONCLUSION

This project has successfully designed and implemented a smart plant pot system with the following main functions: (1) measure soil moisture using a sensor and ESP32, (2) display moisture value and supplementary information on a 2.4-inch TFT via an LVGL interface, and (3) alert when soil moisture is at a dangerous level (too dry or too wet) through interface changes and optional LED/buzzer. The system is suitable as a demo product or a platform for extending research on automatic irrigation and small-scale agricultural IoT.

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Report for the Smart Plant Pot project – ESP32, TFT 2.4”, soil moisture sensor, LVGL interface, and moisture alert.